



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-08	Team No-25	Project Title-Vehicle Parking Finder & Booking App
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1. Abstract

The Vehicle Parking Finder & Booking App is a smart parking management system designed to help vehicle owners find, check, and book available parking spaces quickly and conveniently. Finding a suitable parking space in busy areas such as shopping malls, colleges, offices, hospitals, and commercial locations can be time-consuming and frustrating. Users often have to drive around searching for vacant spaces, which can lead to traffic congestion, fuel wastage, and unnecessary delays. This project aims to solve these problems by providing a centralized platform for locating and reserving parking spaces.

The application allows users to search for nearby parking areas based on their location and view important information such as parking availability, location, pricing, parking capacity, and operating hours. Users can select a suitable parking slot, choose the required time duration, and book the slot in advance. The system can maintain booking details and update parking availability whenever a slot is reserved or released. This helps users plan their parking before reaching their destination and reduces the uncertainty associated with finding a vacant space.

The system also provides useful features for parking administrators or owners. They can add and manage parking locations, monitor available and occupied slots, update parking charges, and manage user bookings. A structured database can be used to store user information, parking locations, slot details, and booking records securely. The application can also provide booking confirmation and maintain a history of previous reservations.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030016	Cherukuri Hamsini	
2520030116	Hasini Reddy	

3. Frameworks / Platforms / Software Used

- Frontend: HTML, CSS, JavaScript
- Framework: React.js for developing an interactive and responsive user interface
- Backend: Node.js with Express.js for handling server-side operations and APIs
- Database: MySQL for storing user, parking, slot, and booking information
- Development Environment: Visual Studio Code (VS Code)
- Version Control: Git and GitHub for managing and tracking project code
- API Testing: Postman for testing backend APIs
- Maps & Location: Google Maps API for displaying parking locations and navigation
- Web Browser: Google Chrome for testing and running the application
- Design/Prototyping: Figma for designing the application interface and user experience

Guide Name and Signature (remarks if any) _____